3-17-26 (WEEK 4)

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EXAMPLE

A prof is grading HW while watching Man City vs Real Madrid.
If Real Madrid wins, prof is happy, and students pass with probability 80%
If they don't, prof is sad, and students pass with probability 50%
Say Real Madrid wins with probability 67%, what is the probability a student will pass?

Solution

Let $S = \left\{ (x,y) \;\middle|\; \begin{array}{l} x = \begin{cases} 1, & \text{if Madrid win} \\ 0, & \text{o/w} \end{cases} \\ y = \begin{cases} 1, & \text{if student passes} \\ 0, & \text{o/w} \end{cases} \end{array} \right\}$

Let $F_1 = \{\text{Real Madrid win}\}$
$F_2 = \{\text{Real Madrid lose}\}$

Then, $S = F_1 \sqcup F_2$

Let $E = \{\text{student passes}\}$
Then, $\mathbb{P}(F_1) = 0.67 \Rightarrow \mathbb{P}(F_2) = 1 - \mathbb{P}(F_1) = 0.33$
We also have $\mathbb{P}(E|F_1) = 0.8$, $\mathbb{P}(E|F_2) = 0.5$

By the Law of Total Probability, $\mathbb{P}(E) = \mathbb{P}(E|F_1)\mathbb{P}(F_1) + \mathbb{P}(E|F_2)\mathbb{P}(F_2) = 0.8(0.67) + 0.5(0.33)$

THEOREM (BAYES' RULE)

Let $(S, \mathbb{P})$ be a probability space, and suppose we have a partition $S = F_1 \cup F_2 \cup \ldots \cup F_n$ and an event $E \subseteq S$.
Then, $\forall\, 1 \leq i \leq n$,

$$\mathbb{P}(F_i|E) = \frac{\mathbb{P}(E|F_i)\mathbb{P}(F_i)}{\sum_{j=1}^{n} \mathbb{P}(E|F_j)\mathbb{P}(F_j)}$$

Proof

$$\mathbb{P}(F_i|E) \overset{\text{by def}}{=} \frac{\mathbb{P}(F_i \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E \cap F_i)}{\mathbb{P}(E)} \overset{\text{by def}}{=} \frac{\mathbb{P}(E|F_i)\mathbb{P}(F_i)}{\mathbb{P}(E)} \overset{\text{law of total probability}}{=} \frac{\mathbb{P}(E|F_i)\mathbb{P}(F_i)}{\sum_{j=1}^{n} \mathbb{P}(E|F_j)\mathbb{P}(F_j)} \quad \square$$

EXAMPLE

A student passes their HW. What is the probability Real Madrid won?

Solution

We want $\mathbb{P}(F_1|E)$.

By Bayes', $\mathbb{P}(F_1|E) = \dfrac{\mathbb{P}(E|F_1)\mathbb{P}(F_1)}{\mathbb{P}(E|F_1)\mathbb{P}(F_1) + \mathbb{P}(E|F_2)\mathbb{P}(F_2)} = \dfrac{0.8(0.67)}{0.8(0.67) + 0.5(0.33)}$

EXAMPLE

COVID testing.
Say 1% of the population has COVID, and a rapid test is 95% accurate.
A random person tests positive for COVID. What is the probability they actually have COVID?

Events: $E = \{\text{have COVID}\}$, $F = \{\text{test positive}\}$
Goal: $\mathbb{P}(E|F) = ?$
Given: $\mathbb{P}(F|E) = 0.95$, $\mathbb{P}(E) = 0.01 \Rightarrow \mathbb{P}(E^c) = 0.99$, $\mathbb{P}(F|E^c) = \mathbb{P}(\text{positive} \mid \text{no COVID}) = 0.05$

By Bayes' Rule, $\mathbb{P}(E|F) = \dfrac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c)} = \dfrac{0.95(0.01)}{0.95(0.01) + 0.05(0.99)} \approx 0.16$!!

Best way to think about it: If we eradicated COVID, then $\mathbb{P}(\text{COVID} \mid \text{positive}) = 0$, so since only 1% of people have COVID, the probability is still low

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PROBABILITY

City has 3000000 people, and DNA testing is 99.99% accurate.
If a random person gets a match, what is the probability they are the criminal?

Solution

$E = \{\text{have the right person}\}$, $F = \{\text{test gives a match}\}$

Goal: $\mathbb{P}(E|F)$

Given: $\mathbb{P}(E) = \frac{1}{3000000} \Rightarrow \mathbb{P}(E^c) = \frac{2999999}{3000000}$

$\mathbb{P}(F|E) = 0.9999$, $\mathbb{P}(F|E^c) = 0.0001$

By Bayes' Rule, $\mathbb{P}(E|F) = \dfrac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F|E)\mathbb{P}(E) + \mathbb{P}(F|E^c)\mathbb{P}(E^c)} = \dfrac{0.9999\left(\frac{1}{3000000}\right)}{0.9999\left(\frac{1}{3000000}\right) + 0.0001\left(\frac{2999999}{3000000}\right)} \approx 0$!!

INDEPENDENCE

DEFINITION

Given a probability space $(S, \mathbb{P})$, we say two events are independent if $\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$

OBSERVATION

If $\mathbb{P}(F) > 0$, this is equivalent to $\underbrace{\mathbb{P}(E|F)}_{\frac{\mathbb{P}(E\cap F)}{\mathbb{P}(F)}} = \underbrace{\mathbb{P}(E)}_{\frac{\mathbb{P}(E)\mathbb{P}(F)}{\mathbb{P}(F)}}$

EXAMPLE

Roll two fair die.
Let $E = \{\text{first roll is odd}\}$, $F = \{\text{second roll is even}\}$

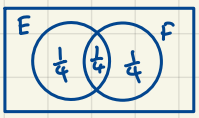
In other words, $S = [6] \times [6]$, $E = \{1, 3, 5\} \times [6]$, $F = \{2, 4, 6\} \times [6]$
Notice, $|S| = 6(6) = 36$, $|E| = 3(6) = 18$, $|F| = 6(3) = 18$
$\therefore \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{1}{2}$, $\mathbb{P}(F) = \frac{|E|}{|S|} = \frac{1}{2}$

Then, $E \cap F = \{1, 3, 5\} \times \{2, 4, 6\} \Rightarrow |E \cap F| = 9 \Rightarrow \mathbb{P}(E \cap F) = \frac{|E \cap F|}{|S|} = \frac{1}{4} = \mathbb{P}(E)\mathbb{P}(F)$
$\therefore$ By def, E and F are independent.

REMARK

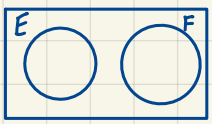
Disjoint $\not\Leftrightarrow$ Independence

Indep:



$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$ ✓

NOT indep:



$\mathbb{P}(E \cap F) = 0 \neq \mathbb{P}(E)\mathbb{P}(F)$

PROPERTIES

① If E, F are disjoint/mutually exclusive, then they are independent $\Leftrightarrow \mathbb{P}(E) = 0$ or $\mathbb{P}(F) = 0$
($\because \mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$)

② If $E \subseteq F$ and they are independent, then $\mathbb{P}(E) = 0$ or $\mathbb{P}(F) = 1$
By indep, $\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$. Yet, $E \subseteq F \Rightarrow E \cap F = E$. $\therefore \mathbb{P}(E) = \mathbb{P}(E)\mathbb{P}(F)$

PROPOSITION

If E and F are independent, then so are E and F^c

Proof

$\mathbb{P}(E) = \mathbb{P}(E \cap S) = \mathbb{P}((E \cap F) \sqcup (E \cap F^c)) = \mathbb{P}(E \cap F) + \mathbb{P}(E \cap F^c)$

As E and F: indep, $\mathbb{P}(E) = \mathbb{P}(E)\mathbb{P}(F) + \mathbb{P}(E \cap F^c) \Rightarrow \mathbb{P}(E \cap F^c) = \mathbb{P}(E)[1 - \mathbb{P}(F)] = \mathbb{P}(E)\mathbb{P}(F^c)$ $\square$

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EXAMPLE

Roll two fair die.

$E = \{\text{first roll is odd}\}$, $F = \{\text{second roll is even}\}$, $G = \{\text{sum is even}\}$

Notice, $\mathbb{P}(G) = \frac{|G|}{|S|} = \frac{18}{36} = \frac{1}{2}$.

Recall, E and F are indep.

Claim: E, G: indep

Proof

$E \cap G = \{1,3,5\} \times \{1,3,5\} \Rightarrow |E \cap G| = 3(3) = 9 \Rightarrow \mathbb{P}(E \cap G) = \frac{|E \cap G|}{|S|} = \frac{1}{4} = \mathbb{P}(E)\mathbb{P}(G)$ $\square$

Similarly F, G: indep

HOWEVER, $\mathbb{P}(E \cap F \cap G) = 0 \neq \mathbb{P}(E)\mathbb{P}(F)\mathbb{P}(G)$!!!

DEFINITION

Given a sequence of events $E_1, E_2, E_3, \ldots$, we say they are mutually independent if for any finite set $I \subseteq \mathbb{N}$, $\mathbb{P}(\bigcap_{i \in I} E_i) = \prod_{i \in I} \mathbb{P}(E_i)$

REMARK

Pairwise indep ("each pair is indep")

$\not\Uparrow$ $\Uparrow$

Mutual indep

EXAMPLE

Consider Inclusion-Exclusion for independent events

Let $E_1, \ldots, E_n$ be mutually independent

Then,
$$\begin{aligned}
\mathbb{P}(E_1 \cup E_2 \cup \ldots \cup E_n) &= \sum_{\phi \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}(\bigcap_{i \in I} E_i) \quad \text{(by incl-excl)} \\
&= \sum_{\phi \neq I \subseteq [n]} (-1)^{|I|+1} \prod_{i \in I} \mathbb{P}(E_i) \quad \text{(by indep)} \\
&= -\sum_{\phi \neq I \subseteq [n]} \prod_{i \in I} (-\mathbb{P}(E_i)) \\
&= 1 - \sum_{I \subseteq [n]} \prod_{i \in I} (-\mathbb{P}(E_i)) \\
&= 1 - \prod_{i=1}^{n} (1 - \mathbb{P}(E_i))
\end{aligned}$$

multiplying $\prod_{i \in \phi} \to 1$

$I = \{\text{factors where we choose the } -\mathbb{P}(E_i) \text{ term}\}$

Alt proof

By de Morgan's Law, $(\bigcup_{i=1}^{n} E_i)^c = \bigcap_{i=1}^{n} E_i^c$

$\therefore \mathbb{P}(\bigcup_{i=1}^{n} E_i) = 1 - \mathbb{P}((\bigcup_{i=1}^{n} E_i)^c) = 1 - \mathbb{P}(\bigcap_{i=1}^{n} E_i^c) \overset{\text{indep}}{=} 1 - \prod_{i=1}^{n} \mathbb{P}(E_i^c) = 1 - \prod_{i=1}^{n} (1 - \mathbb{P}(E_i))$ $\square$